



Balance conditions to cancel out  
 $R_{x0}$  we set

$$K = \frac{R_2}{R_1} = \frac{R_3}{R_{x0}}$$

Let's derive the equation

$$V_o = (V^+ - V^-) \longrightarrow V_o = V_s \left( \frac{R_x}{R_3 + R_x} - \frac{R_2}{R_1 + R_2} \right) \quad R_2 = R_{x0}$$

$$R_x = R_{x0} + \Delta R_x$$

$$V_o = V_s \left( \frac{R_{x0} + \Delta R_x}{R_3 + R_{x0} + \Delta R_x} - \frac{R_{x0}}{R_1 + R_{x0}} \right) \quad K = \frac{R_2}{R_1} = \frac{R_3}{R_{x0}}$$

$$V_o = V_s \left( \frac{R_{x0} + \Delta R_x}{R_{x0}(K+1) + \Delta R_x} - \frac{R_{x0}}{R_{x0}(K+1)} \right) \quad \text{thus } R_{x0} \cdot K = R_1$$

$$V_o = V_s \cdot \left( \frac{(\cancel{R_{x0}} + \Delta R_x)(K+1) - \cancel{R_{x0}}(K+1) + \Delta R_x}{(R_{x0}(K+1) + \Delta R_x)(K+1)} \right) \quad \Delta R_x(K+1)$$

$$V_o = V_s \cdot \frac{K \Delta R_x}{(R_{x0}(K+1) + \Delta R_x)(K+1)} \quad \text{if } \Delta R_x \ll R_{x0}$$

$$V_o = \frac{K \Delta R_x}{R_{x0}(K+1)^2} \cdot V_s \quad \text{and } \Delta R_x = \alpha R_{x0} \Delta T$$

$$V_o = V_s \cdot \frac{K \alpha R_{x0} \Delta T}{R_{x0}(K+1)^2}$$

Temperature coefficient.

Note that  $V_o \sim \Delta T$  by a coeff.

$$V_o = V_s \cdot \frac{K \alpha}{(K+1)^2} \cdot \Delta T \quad \text{hence there is a linear relationship depending on } \Delta T$$

Let's bring it back to  $R_{x0} \cdot K = R_1$

$$V_o = V_s \cdot \frac{R_{x0} \cdot R_1}{(R_1 + R_{x0})^2} \cdot \alpha \cdot \Delta T$$

Since we would like an output dependent only on  $\Delta R_{x0}$  and not  $R_{x0}$  we must

$$\frac{R_2}{R_3} = \frac{R_1}{R_{x0}}$$

And if we wanna maximize the  $V_{out}$  with its derivative, we'll find that it is going to lead us to

$$R_1 = R_2 = R_3 = R_{x0}$$

Then

$$V_0 = V_s \cdot \frac{R_{x0} \cdot R_1}{(R_1 + R_{x0})^2} \cdot \alpha \cdot \Delta T$$

$$\rightarrow V_0 = V_s \cdot \frac{\cancel{R_{x0}}}{4\cancel{R_{x0}}^2} \cdot \underbrace{R_{x0} \cdot \alpha \cdot \Delta T}_{\Delta R_{x0}}$$

$$V_0 = \frac{1}{4} \cdot \frac{\Delta R_{x0}}{R_x} \cdot V_s$$